

Predicting first onset of suicide attempt among children with suicidal ideation or non-suicidal self-injury using machine learning: a prospective population-based cohort study

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Abstract

While children with suicidal ideation or non-suicidal self-injury (NSSI) are at high risk of suicide, most do not attempt suicide. This study aims to identify predictors of first suicide attempts among children with suicidal thoughts or NSSI. We utilized longitudinal data from the Adolescent Brain and Cognitive Development study to examine a broad range of risk factors in children reporting suicidal ideation ($N = 344$, mean age = 9.93) or NSSI ($N = 261$, mean age = 9.88) at baseline, predicting first attempt of suicide during a 4-year follow-up. Predictive models were developed through bivariate feature selection followed by Discrete-Time Random Survival Forest machine learning, identifying predictors of subsequent suicide attempts. During the follow-up, 40 of 344 children (11.6%) with suicidal ideation, and 32 of 261 children (12.3%) with NSSI at baseline initiated suicide attempts. For suicidal ideation, risk for attempting suicide increased with caregiver-reported NSSI, witnessing domestic violence, severity of suicidal thoughts, being female, online social screen use, and less parental supervision. For NSSI, risk of suicide attempt increased with witnessing domestic violence, anxiety disorders, caregiver-reported NSSI, being female, and disruptive, impulse control, and conduct disorders. Our findings shed light on the development of population-based suicide prevention strategies for children.

Key words: suicide attempt; suicidal ideation; non-suicidal self-injury; initiation; transition; prediction.

Introduction

Suicide is a leading cause of death among children, highlighting a significant public health concern globally.¹ However, there is a dearth of high-quality, large-scale research focused on children.² Despite the paucity of research, late childhood, delineated as encompassing ages ten to twelve years, emerges as a crucial developmental stage for the initiation of suicidal behaviors. A nationally representative study in the United States found a negligible incidence of suicide attempts before the age of 10 years, followed by an increase starting at age 10 years.³ Adding to this concern, recent data from the Centers for Disease Control and Prevention identify suicide as the second leading cause of death among 10- to 14-year-olds in the United States.⁴ Although children with suicidal ideation and non-suicidal self-injury (NSSI) are at high risk for attempting suicide,⁵ only a small percentage report a lifetime suicide attempt (29.0% and 11.8%, respectively).^{6,7} Therefore, it is crucial to identify which children within these high-risk groups

are likely to attempt suicide subsequently. This study aims to address this issue by exploring the factors that predict suicide attempts among children with suicidal ideation or engaging in NSSI, thereby informing both suicide screening and prevention.

Suicide capability has been proposed to be a key factor in the transition from thoughts to action in several theoretical models.⁸ The Interpersonal–Psychological Theory of Suicide (IPTs) proposes that acquired capability, manifested as lowered fear of death and increased pain tolerance intensified through exposure to painful and provocative events, plays a critical role in translating suicidal ideation to action.^{9,10} Expanding on acquired capability, the Integrated Motivational–Volitional (IMV) model introduces additional volitional factors, which are characterized as those that could contribute to the enactment of suicidal actions, such as impulsivity and access to lethal means.¹¹ Further refining such factors, the Three-Step Theory (3ST) posits that dispositional, acquired, and practical capacities, reflecting the influences

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of dispositional temperaments, learned experiences of pain and violence, and knowledge and accessibility of lethal means collectively contribute to the capacity to initiate suicide attempts among strong ideators.^{12,13} Of note, the transition from NSSI to suicide attempts has received relatively less attention. Hamza and colleagues proposed two possible roles of its influence relevant to suicidality: one where NSSI acts as a precursor of suicide attempts, driven by intrapersonal distress that may escalate NSSI to suicidal self-injury and another where NSSI contributes to increased acquired capability for suicide by desensitizing the individual to self-harm, which, when combined with suicidal ideation, facilitates suicide attempts.¹⁴

To our knowledge, we conducted the first high-quality large-scale study that has examined the factors prospectively associated with the transition from suicidal ideation to actions or from NSSI to suicidal ones using a population-based sample in children. Relevantly, past research has contributed valuable insights into factors associated with these transitions in other age groups. However, the predominant methodological designs of these studies remain cross-sectional or retrospective.¹⁵ Only seven studies adopted a prospective design, but they are beset with limitations.^{16–22} First, many of these studies lacked representativity by not using population-based samples.^{16–18} Moreover, some failed to distinguish between initial and recurrent suicide attempts, blurring predictors of initiation and maintenance.^{16–20} Additionally, some studies examined a narrow range of risk factors,^{17,21} hindering a comprehensive characterization of children at risk for initiating suicide attempts. Furthermore, several relied on self-report rather than diagnostic interviews for the assessment of NSSI, suicidal ideation, and suicide attempts, which may compromise the reliability or validity of these assessments.^{20,22} Last but not least, all these previous studies have relied on traditional statistical methods (eg, logistic regression), which have limited their ability to explore non-linear complex patterns among variables.^{16–22} To our knowledge, there is only one high-quality, large-scale study that examined the factors prospectively associated with the transition from suicidal ideation or NSSI to suicide attempts, but in adolescence and early adulthood and using traditional statistical methods.²² The knowledge gap exemplified through the lack of research on prospective predictors of transition to suicide attempts in children is critical because age could substantially influence the nature of risk factors involved. For example, ongoing substance use might be an important risk factor for adolescents, but not for younger children, as past research shows that the base rate of ongoing substance use among children is extremely low (eg, 0%–0.4% of daily smoking for sixth and seventh grade children).²³ Moreover, new putative risk factors have emerged over the past few years that have been overlooked in previous studies, such as problematic screen use.²⁴ Additionally, for younger children, caregiver reports of risk may also be important as children may be less insightful compared to adolescents, given the developmental and cognitive differences.²⁴

Our primary goal is to elucidate the putative risk factors predicting the transition from suicidal ideation or NSSI to a first attempt of suicide among children, thereby informing the development of targeted screening, prevention, and intervention approaches for children at risk. We utilized the Adolescent Brain and Cognitive Development (ABCD) study, taking the unprecedented opportunity to examine childhood suicidal thoughts and behaviors through the largest nationally representative, population-based cohort of 9- to 10-year-olds residing in the United States. Specifically, we employed a machine learning approach to capture complex patterns among a comprehensive

list of over 60 potential risk factors identified through empirical findings and theoretical models for suicide attempts. A detailed review of these risk factors and their association with suicide attempts is available in [Table S1](#). Informed by theoretical models, we expect that among children experiencing suicidal ideation, severity of ideation and risk factors indicative of heightened suicidal capability or volition, such as NSSI and traumatic events, may be predictive of subsequent suicide attempts. Among children engaging in NSSI, risk factors indicating suicidal capability or presence of baseline suicidal ideation may be critical predictors of future suicide attempts. Additionally, predictive models will be derived from these factors and their predictive accuracy will be assessed to validate the predictive power.

Method

Our study protocol, including analysis plans, was pre-registered on the Open Science Framework (available at osf.io/8rqgm).

Participants

For the current study, data were sourced from the ABCD Study's Curated Annual Release 5.0. This release compiles longitudinal records collected from September 1, 2016, to January 15, 2022, with baseline data from 11 868 participants, followed subsequently with a yearly interval for up to four years. Written informed consent was obtained from all children and their caregivers. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2013. All procedures involving human subjects/patients were approved by the central institutional review board at the University of California, San Diego, along with local IRB approvals from each respective site (approval number: #160091). Further information about study design and recruitment is accessible at the ABCD Study website (<https://wiki.abcdstudy.org/release-notes/start-page.html>).

The current investigation incorporated data across all five assessment waves, from baseline to the 4-year follow-up. Participants included in the current study completed interview assessment of suicide attempts across all five assessments. Among them, children who reported suicidal ideation or engaging in NSSI but without suicide attempts at baseline were selected. Given the limitations in caregiver-reported suicide data,²⁵ this study primarily relied on children reports for assessments of suicidal ideation, suicide attempts, and NSSI. This selection resulted in two final samples for predicting initiation of suicide attempts: the suicidal ideation group ($n = 344$), with mean age of 9.93 years ($SD = 0.63$) and comprising 206 males (59.9%); and the NSSI group ($n = 261$), with a mean age of 9.88 years ($SD = 0.61$) and comprising 177 males (67.8 %). The detailed selection and exclusion criteria are delineated in [Figure 1](#). Demographic characteristics of the two samples are shown in [Table 1](#).

Measurements

The study variables, including their names/labels in the ABCD study, any relevant recoding, the interpretation of rating scales/scores, and examples of items, are provided in detail in [Table S2](#). For a comprehensive description of the measures utilized in the ABCD study, foundational literature including Barch et al. and Zucker et al. can be referenced.^{26,27}

Outcome: suicide attempts

Suicide attempts were assessed using the Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS-5).²⁸ We

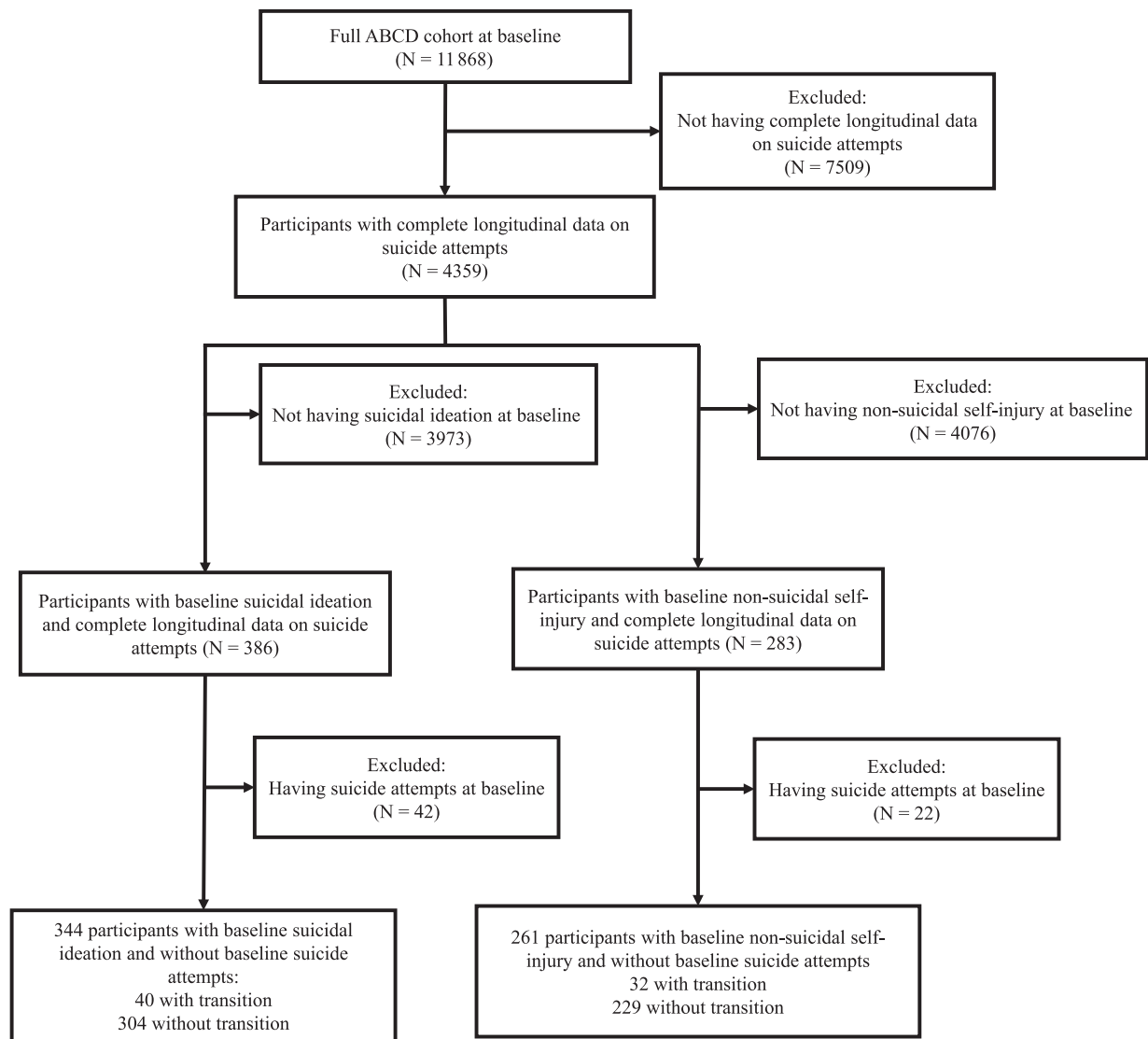


Figure 1. Flow chart of attrition for the current study.

aggregated reports of present or past interrupted, aborted, and completed suicide attempts into one category of “any suicide attempt.” For the analysis of transitions from baseline suicidal ideation or NSSI to first attempt of suicide, participants with any suicide attempt during any follow-up but not at baseline are categorized as the “transition group,” while those without suicide attempt are categorized throughout as the “no transition group.”

Cluster 1: demographics

We collected demographic details of the study participants, which included their age, family income, parent education level, sex assigned at birth, and race, all of which were provided in the ABCD Demographics Survey.

Cluster 2: suicidal ideation characteristics

Suicidal ideation characteristics were evaluated using KSADS-5.²⁸ Various forms of present or past suicidal ideation at baseline, including passive, nonspecific active, active with a specific method, active with an intent, and active with a plan, were aggregated as “baseline suicidal ideation.” Suicidal ideation severity was coded from 0 to 5, indicating the absence of suicidal

ideation, passive suicidal ideation, nonspecific active suicidal ideation, suicidal ideation with a method, suicidal ideation with an intent, and suicidal ideation with a plan. Suicidal ideation currency was coded based on the presence of suicidal ideation at baseline (0, no present suicidal ideation; 1, present suicidal ideation). We included both caregiver and child reports as risk factors to account for possible differences in reporting styles, as each may provide unique insights with distinct predictive value.²⁹

Cluster 3: NSSI characteristics

Similarly, NSSI was also measured using KSADS-5 and both past and present NSSI were aggregated as “baseline NSSI.” Both caregiver and child reports were included as risk factors.²⁸

Cluster 4: psychopathology

The severities of internalizing and externalizing symptoms were assessed through the caregiver-reported Child Behavior Checklist (CBCL).³⁰ Mental health service use was assessed using caregiver-reported KSADS-5. The presence of lifetime psychiatric disorders was also identified using the KSADS-5, based on self-reports and caregiver reports of the child’s current or past disorders at baseline. Information on depressive, anxiety, bipolar, and

Table 1. Demographic characteristics of participants at baseline, United States, 2016-2022.

	Transition from suicidal ideation to suicide attempts (N = 344)		Transition from non-suicidal self-injury to suicide attempts (N = 261)	
	M	SD	M	SD
Age	9.93	0.63	9.88	0.61
	N	%	N	%
Sex				
Male	206	59.9	177	67.8
Female	138	40.1	84	32.2
Race				
White	211	61.3	159	60.9
Asian	27	7.8	27	10.3
Hispanic	61	17.7	48	18.4
African American	10	2.9	5	1.9
Other	35	10.2	22	8.4
Parental education				
Without high school diploma	10	2.9	4	1.5
High school graduate without college education	27	7.8	21	8.0
Some college education	95	27.6	83	31.8
Degree from 4-year college or more	212	61.6	152	58.2
Missing	0	0.0	1	0.4
Family income for the past 12 months				
Less than \$5000	4	1.2	7	2.7
\$5000 through \$11 999	8	2.3	10	3.8
\$12 000 through \$15 999	3	0.9	4	1.5
\$16 000 through \$24 999	14	4.1	17	6.5
\$25 000 through \$34 999	11	3.2	10	3.8
\$35 000 through \$49 999	24	7.0	18	6.9
\$50 000 through \$74 999	49	14.2	39	14.9
\$75 000 through \$99 999	56	16.3	35	13.4
\$100 000 through \$199 999	123	35.8	78	29.9
\$200 000 and greater	31	9.0	29	11.1
Missing	21	6.1	14	5.4

related disorders was provided by both caregivers and children, whereas data on other psychiatric disorders were solely provided by caregivers, in accordance with the design of the ABCD database.

Cluster 5: temperaments

Behavioral inhibition and activation were measured using the Behavioral Inhibition & Activation Scale (BIS-BAS),³¹ while impulsivity was assessed using the Urgency, Perseverance, Premeditation, and Sensation Seeking Scale (UPPS),³² with both scales relying on child's self-reports.

Cluster 6: lifestyle factors

The lifestyle factors included three different types of screen time usage and sleep disturbances. Self-reported screen use time on online social, screen watch, and video games was assessed using the adapted Screen Time Questionnaire for children (STQ).³³ Child sleep disturbances were assessed using the Sleep Disturbances Scales for Children (SDS) reported by caregivers.³⁴

Cluster 7: social-environmental factors

Social-environmental factors included family environment, school environment, peer relations, and exposure to traumatic life events. Child reports of family conflict, parental supervision, and caregiver acceptance were assessed using the Conflict Subscale of the Family Environment Scale (FES),³⁵ the Parental Monitoring Scale (PMS),³⁶ and the Acceptance Subscale

from Children's Reports of Parental Behavior Inventory-Short (CRPBI-S),³⁷ respectively. Additionally, caregiver reports of family conflict were collected. Child reports of school disengagement, school involvement, and perceived school environment were assessed using the School Risk & Protective Factors Inventory of the PhenX toolkit (SRPF).³⁸ Caregiver-reported child data on child friendships (ie, having a best friend or regular group of friends, having experiences of bullying) were gathered during the introductory interview of KSADS-5.²⁸ Finally, exposure to traumatic events was evaluated using the posttraumatic stress disorder (PTSD) module of the KSADS-5, with caregiver reports on a range of traumatic experiences faced by the child (eg, car accident, witnessed domestic violence).²⁸

Cluster 8: parental psychopathology

Parents' experiences with various mental health issues (eg, depression, mania, substance use, suicide), as well as issues related to occupation, legal matters, or aggression, were evaluated using the Family History Assessment Module (FHAM),³⁹ based on reports from caregivers.

Cluster 9: perinatal factors

The caregiver-reported Developmental History Questionnaire (DHQ) was used to collect data on a variety of perinatal factors,⁴⁰ including maternal/paternal age at childbirth, maternal medical conditions during pregnancy, obstetric complications at birth, birth weight, substance/medicine use during pregnancy, planned pregnancy or not, premature birth or not.

Statistical analysis

Given the shared characteristics of the two datasets, specifically, their relatively small sample sizes, high dimensionality, and discrete-time structure, we selected the Discrete-Time Random Survival Forest (RSF)⁴¹ algorithm for predicting suicide attempts among children with suicidal ideation or NSSI. RSF extends the conventional Random Forest framework to accommodate right-censored time-to-event data, making it particularly well-suited for survival analysis in the presence of censoring.

Missing values were addressed using mean substitution for continuous variables and the modal category for categorical variables. To reduce noise and mitigate overfitting, we first conducted univariate Cox proportional hazards regression to screen variables for inclusion. To minimize the risk of excluding potentially informative predictors, a liberal threshold of $P < .10$ was adopted. The preliminarily screened variables were used to construct a RSF model with 5000 trees.

To enhance generalizability, 5-fold stratified cross-validation was employed, with oversampling applied to positive samples (ie, initiating suicide attempts) during training to mitigate class imbalance. The survival probability threshold for predicting initiation of suicide attempts at year t was automatically determined in each fold using the Youden index.⁴² In addition, to account for potential dependencies introduced by sibling relationships, a sensitivity analysis was conducted by randomly retaining only one child from each sibling pair (15/4 sibling pairs in the suicidal ideation/NSSI group).

Model performance was assessed using a comprehensive set of metrics, including the area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and the integrated Brier score (IBS).^{43,44} The 95% CIs of AUC were estimated using DeLong's method.⁴⁵

To evaluate the importance of candidate predictors to model performance, variable importance was estimated based on the decrease in AUC resulting from variable permutation, averaged across cross-validation folds to ensure robustness.⁴⁶ To further interpret the model, partial dependence plots were generated for the most influential predictors.⁴⁷

All statistical analyses were conducted using Python version 3.11.9. Model development was implemented using the "lifelines" and "scikit-survival" packages, with evaluation metrics and class imbalance handled by "scikit-learn" and "imbalanced-learn."⁴⁸⁻⁵⁰

Results

Prevalence

Within the cohort of children exhibiting suicidal ideation at baseline ($n = 344$), 40 (11.6%) manifested their first suicide attempt during the following 4 years. Among children who engaged in NSSI at baseline ($n = 261$), 32 (12.3%) initiated a first attempt within the 4-year follow-up period.

Model discrimination

Feature selection identified 28 variables in the suicidal ideation group and 20 variables in the NSSI group (see [Tables 2 and 3](#)). Based on these predictors, the models demonstrated good discriminative ability over the 4-year follow-up. For the suicidal ideation group, the model achieved an AUC of 0.72 (95% CI, 0.62-0.79), with an IBS score of 0.05. For the NSSI group, the AUC was 0.70 (95% CI, 0.63-0.82), with an IBS score of 0.07. Time-dependent receiver operating characteristic curves are shown in [Figure 2](#).

After employing Youden index to determine thresholds, the suicidal ideation model yielded an accuracy of 0.71, sensitivity of 0.93, specificity of 0.71, PPV of 0.17, and NPV of 0.99. Similarly, the NSSI model yielded an accuracy of 0.72, sensitivity of 0.79, specificity of 0.71, PPV of 0.17, and NPV of 1.00. Sensitivity analyses yielded highly similar results ([Appendix S1](#) and [Figure S1](#)), confirming the stability of these findings.

Overall, these findings demonstrate that machine learning models achieved good discriminative performance when separately applied to the suicidal ideation and NSSI groups, and may help delineate high-risk subgroups for suicide attempt prediction.

Predictor importance

[Figure 3](#) depicts the ten most influential variables for each prediction model. Partial dependence plots are shown in [Figure 4](#). Sensitivity analyses yielded highly similar sets of important predictors across both groups, with details provided in [Figure S2](#).

Among children with suicidal ideation, the strongest predictor was caregiver-reported non-suicidal self-injury, followed by exposure to domestic violence, greater severity of self-reported suicidal ideation, female sex, increased online social screen time, and less parental supervision. Among children engaging in NSSI, exposure to domestic violence was the most significant predictor, followed by self-reported anxiety disorders, caregiver-reported non-suicidal self-injury, female sex, behavioral activation, and caregiver-reported disruptive, impulse-control, and conduct disorders.

Taken together, these top predictors encompass a broad spectrum of psychosocial and clinical domains, underscoring the complex and multifactorial nature of the early transition from suicidal ideation or NSSI to suicide attempts.

Discussion

To our knowledge, by leveraging ABCD database, this is the first prospective population-based study to explore the risk factors associated with the initiation of suicide attempts among children with suicidal ideation or NSSI using a machine learning approach. For 344 children who reported suicidal ideation at baseline, 40 (11.6%) initiated suicide attempts during the following four years. For 261 children who engaged in NSSI at baseline, 32 (12.3%) initiated suicide attempts during the subsequent four years. These prevalence rates are highly consistent with the 12% transition rates previously observed in adolescents²². This consistency may indicate a uniform pattern in these transitions across different age groups.

Our prediction models revealed robust and important predictors for the transition from suicidal ideation or NSSI to suicide attempts, with being assigned female at birth standing out as a significant factor in both groups, echoing prior research that associates females with a higher risk and a faster pace of transition.^{16,51} Although some studies on child suicidality have reported the opposite trend, with males exhibiting a higher risk towards suicidality,²⁴ this inconsistency might reflect age-related differences, as older female children may be more likely to encounter greater interpersonal stress,⁵² and undergo pubertal development, introducing biological and psychological vulnerabilities,⁵³ a possibility awaiting future investigation. Collectively, these findings suggest that female children should receive particular attention in suicide prevention efforts, especially in high-risk groups.

Furthermore, although largely exploratory in nature, several key predictors identified in the current study align with the

Table 2. Descriptive and univariate Cox proportional hazards regression analyses of baseline risk factors comparing transition group and no transition group when predicting transitions from suicidal ideation to suicide attempts, United States, 2016-2022.

	Transition from suicidal ideation to suicide attempts				Missing (%)
	Transition group (N = 40)	No transition group (N = 304)	P value	HR [95% CI]	
	M (SD)	M (SD)/%			
Demographics					
Age	119.93 (7.13)	119.12 (7.58)	0.51	1.01 [0.97-1.06]	0.0
Family income	7.12 (2.18)	7.71 (2.01)	0.03	0.86 [0.75-0.98]	6.1
Parent education level	16.95 (2.07)	17.20 (2.44)	0.52	0.96 [0.85-1.08]	0.0
Female sex	62.5%	37.2%	0.00	2.59 [1.36-4.91]	0.0
Race	..	..	..	..	0.0
White	55.0%	62.2%	..	1 (ref)	..
Black	7.5%	7.9%	0.94	0.96 [0.29-3.10]	..
Hispanic	25.0%	16.8%	0.19	1.61 [0.79-3.29]	..
Asian	5.0%	2.6%	0.46	1.70 [0.41-7.04]	..
Other	7.5%	10.5%	0.54	0.69 [0.21-2.24]	..
Suicidal ideation characteristics					
Suicidal ideation severity (caregiver report)	0.66 (1.04)	0.35 (0.76)	0.03	1.39 [1.04-1.85]	0.9
Suicidal ideation severity (child report)	2.08 (1.31)	1.60 (0.90)	0.00	1.43 [1.12-1.82]	0.0
Suicidal ideation currency (caregiver report)	6.5%	4.9%	0.89	1.11 [0.27-4.58]	0.9
Suicidal ideation currency (child report)	35.0%	21.4%	0.05	1.91 [1.00-3.65]	0.0
Non-suicidal self-injury					
Non-suicidal self-injury (child report)	45.0%	25.0%	0.01	2.34 [1.25-4.35]	0.0
Non-suicidal self-injury (caregiver report)	29.3%	9.6%	0.00	3.29 [1.64-6.59]	0.9
Psychopathology					
Internalizing symptoms	55.50 (12.55)	52.15 (10.99)	0.07	1.03 [1.00-1.05]	0.0
Externalizing symptoms	54.78 (10.71)	48.77 (10.58)	0.00	1.05 [1.02-1.08]	0.0
Mental health service use	42.5%	27.4%	0.04	1.92 [1.03-3.59]	0.6
Depressive disorders (child report)	27.5%	14.5%	0.03	2.12 [1.06-4.24]	0.0
Anxiety disorders (child report)	15.0%	12.8%	0.70	1.19 [0.50-2.83]	0.0
Bipolar and related disorders (child report)	30.0%	23.0%	0.31	1.42 [0.72-2.79]	0.0
Depressive disorders (caregiver report)	17.5%	9.1%	0.07	2.11 [0.93-4.78]	0.9
Anxiety disorders (caregiver report)	50.3%	38.8%	0.15	1.57 [0.84-2.92]	0.9
Bipolar and related disorders (caregiver report)	19.0%	6.6%	0.01	3.05 [1.35-6.90]	0.9
Schizophrenia spectrum and other psychotic disorders (caregiver report)	6.3%	3.0%	0.42	1.80 [0.43-7.47]	0.9
Obsessive-compulsive and related disorders (caregiver report)	17.8%	14.0%	0.51	1.31 [0.58-2.97]	0.9
Trauma- and stressor-related disorders (caregiver report)	8.8%	9.3%	0.76	0.83 [0.26-2.70]	0.9
Disruptive, impulse-control, and conduct disorders (caregiver report)	35.0%	22.5%	0.07	1.83 [0.96-3.51]	0.9
Temperaments					
Behavioral inhibition	11.28 (2.88)	10.53 (3.92)	0.26	1.05 [0.97-1.13]	0.0
Behavioral activation	1.81 (0.59)	1.61 (0.55)	0.04	1.80 [1.03-3.13]	0.0
Impulsiveness	9.20 (1.74)	8.80 (1.52)	0.12	1.17 [0.96-1.43]	0.0
Lifestyle factors					
Online social screen time	1.13 (1.30)	0.63 (0.90)	0.00	1.39 [1.12-1.73]	0.0
Screen watch screen time	3.08 (1.58)	2.61 (1.42)	0.05	1.24 [1.00-1.53]	0.0
Video games screen time	1.48 (0.99)	1.42 (1.02)	0.73	1.05 [0.78-1.42]	0.0
Sleep disturbance	39.45 (7.42)	38.10 (9.53)	0.40	1.01 [0.98-1.04]	0.0
Social-environment factors					
Family conflict (child-report)	3.15 (2.11)	2.63 (2.14)	0.15	1.11 [0.97-1.27]	0.3
Family conflict (caregiver-report)>	2.70 (2.00)	2.60 (1.97)	0.73	1.03 [0.88-1.20]	0.0
Parental supervision	4.08 (0.66)	4.32 (0.53)	0.01	0.50 [0.30-0.83]	0.6
Primary caregiver's acceptance	2.67 (0.37)	2.73 (0.34)	0.26	0.63 [0.29-1.40]	0.3
School disengagement	4.48 (1.77)	3.93 (1.47)	0.03	1.23 [1.02-1.49]	0.3
School involvement	12.18 (3.27)	12.35 (2.42)	0.61	0.97 [0.86-1.10]	0.3
Positive view of school environment	18.65 (3.49)	19.28 (2.92)	0.20	0.94 [0.85-1.03]	0.3
Bullying	50.0%	31.4%	0.03	1.96 [1.05-3.64]	1.5
Best friend	67.8%	78.2%	0.20	0.63 [0.32-1.27]	9.0
Friend group	77.0%	87.4%	0.14	0.55 [0.26-1.20]	4.4
Car accident	10.0%	6.3%	0.73	1.23 [0.38-3.98]	1.2
Accident	9.8%	3.0%	0.09	2.81 [0.87-9.12]	1.2
Witnessed domestic violence	21.3%	6.7%	0.00	3.18 [1.46-6.90]	1.2
Sudden death of loved one	30.3%	26.7%	0.88	1.05 [0.53-2.11]	1.2

(Continued)

Table 2. Continued

	Transition from suicidal ideation to suicide attempts				Missing (%)
	Transition group (N = 40)	No transition group (N = 304)	P value	HR [95% CI]	
	M (SD)	M (SD)/%			
Parental psychopathology					
Parental trouble with job/fights/police problem	24.0%	14.4%	0.10	1.87 [0.89-3.93]	3.5
Parental nerves/nervous breakdown problem	23.0%	14.5%	0.40	1.42 [0.63-3.20]	7.3
Parental mania problem	11.5%	9.3%	0.78	1.18 [0.36-3.84]	6.4
Parental drug use problem	28.0%	11.7%	0.06	2.10 [0.97-4.57]	4.4
Parental depression problem	56.5%	41.3%	0.15	1.58 [0.85-2.94]	6.7
Parental alcohol problem	31.8%	16.9%	0.11	1.78 [0.87-3.64]	4.1
Parent hospitalized due to emotional/mental problem	31.0%	14.0%	0.06	2.02 [0.96, 4.24]	6.1
Parent been to a doctor or counselor due to emotional/mental problem	53.0%	51.3%	0.87	1.05 [0.57-1.96]	6.4
Parent attempted or completed suicide	14.8%	10.4%	0.46	1.48 [0.53-4.15]	7.3
Perinatal factors					
Maternal medical conditions during pregnancy	0.79 (1.10)	0.71 (0.97)	0.57	1.09 [0.80-1.49]	6.7
Obstetric complications	0.48 (0.88)	0.47 (0.95)	0.92	1.02 [0.72-1.43]	9.0
Birth weight	7.69 (4.47)	7.49 (4.39)	0.74	1.01 [0.94-1.09]	7.8
Paternal age at child's birth	29.48 (7.45)	32.24 (7.11)	0.02	0.94 [0.89-0.99]	4.1
Maternal age at child's birth	27.20 (6.21)	29.74 (5.93)	0.01	0.93 [0.88-0.98]	2.3
Planned pregnancy	48.8%	66.5%	0.02	0.49 [0.26-0.91]	3.2
Medication use during pregnancy	17.0%	16.3%	0.88	1.07 [0.45-2.54]	5.5
Maternal alcohol/substance use during pregnancy	16.3%	8.8%	0.10	2.07 [0.87-4.93]	2.6
Premature birth	10.0%	20.0%	0.15	0.47 [0.17-1.33]	1.5

ideation-to-action theoretical framework.⁸ Notably, witnessing domestic violence, a factor suggesting increased acquired capacity for suicide attempts through greater adaptation to violence, was a robust predictor of the transition for both groups. Echoing prior findings that adults who experienced domestic violence are at elevated risk for attempting suicide,⁵⁴ the finding that witnessing alone is enough for children at risk to escalate the risk is quite noteworthy. This finding suggests that incorporating family-level preventive measures might be important for suicide prevention among children. Consistently, engaging in NSSI also

predicted subsequent attempts for both groups, potentially by increasing an individual's acquired capacity for suicide. Moreover, higher behavioral activation in the NSSI group was an important predictor, potentially reflecting a dispositional capacity to act on suicidal urges, echoing Hamza et al.'s proposal that individuals may gradually accumulate suicide capacity over time. Furthermore, for children with suicidal thoughts, severity of suicidal ideation also emerged as a significant risk factor in the progression from suicidal ideation to suicide attempts, which is highly consistent with previous theoretical and empirical

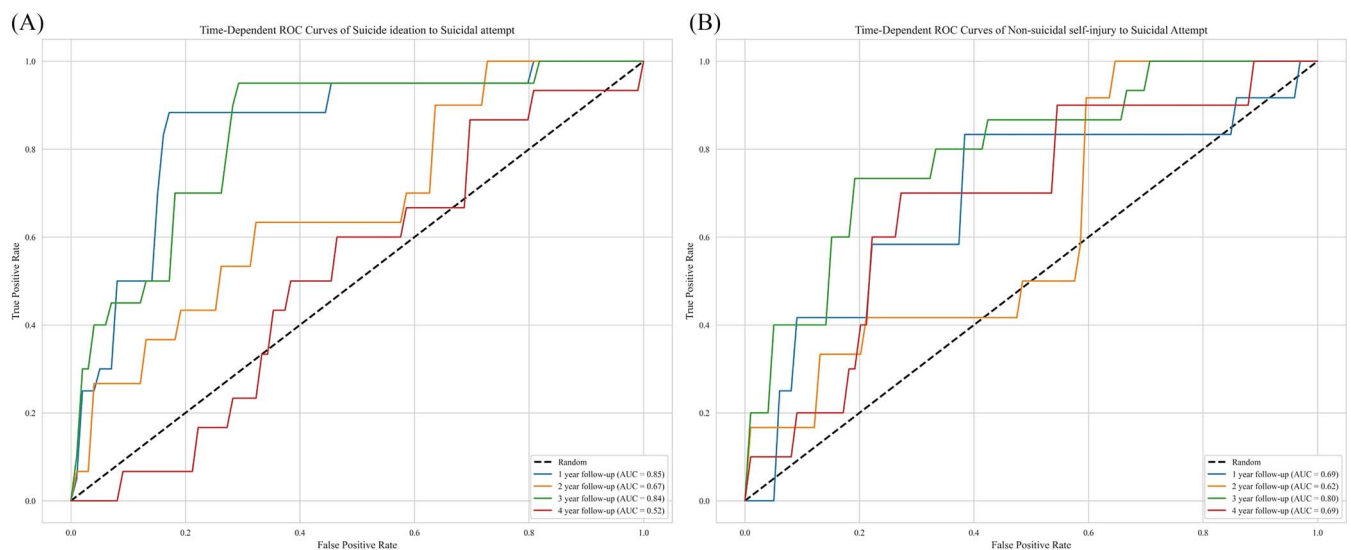


Figure 2. Time-dependent ROC curves for the prediction models. (A) Time-dependent ROC curves for the prediction model of transition from suicidal ideation to suicide attempt. (B) Time-dependent ROC curves for the prediction model of transition from non-suicidal self-injury to suicide attempt.

Table 3. Descriptive and univariate Cox proportional hazards regression analyses of baseline risk factors comparing transition group and no transition group when predicting transitions from non-suicidal self-injury to suicide attempts, United States, 2016-2022

	Transition from non-Suicidal self-Injury to suicide attempts				Missing (%)
	Transition group (N = 32)	No transition group (N = 229)	P value	HR [95% CI]	
	M (SD)	M (SD)			
Demographics					
Age	119.03 (7.00)	118.48 (7.38)	0.67	1.01 [0.96-1.06]	0.0
Family income	6.31 (2.77)	7.48 (2.26)	0.01	0.84 [0.74-0.95]	5.4
Parent education level	16.53 (2.34)	17.18 (2.32)	0.11	0.89 [0.78-1.03]	0.4
Female sex	50.0%	29.7%	0.03	2.20 [1.10-4.40]	0.0
Race	..	..	..	..	0.0
White	65.6%	60.3%	..	1 (ref)	..
Black	0.0%	11.8%	..	..	..
Hispanic	21.9%	17.9%	0.57	1.28 [0.55-2.95]	..
Asian	3.1%	1.7%	0.65	1.59 [0.22-11.62]	..
Other	9.4%	8.3%	0.88	1.10 [0.33-3.61]	..
Suicidal ideation characteristics					
Suicidal ideation severity (caregiver report)	0.61 (1.05)	0.25 (0.66)	0.01	1.59 [1.15-2.19]	1.5
Suicidal ideation severity (child report)	1.28 (1.61)	0.63 (1.11)	0.01	1.35 [1.09-1.67]	0.0
Suicidal ideation currency (caregiver report)	4.4%	3.2%	0.82	1.26 [0.17-9.21]	1.5
Suicidal ideation currency (child report)	15.6%	9.2%	0.23	1.80 [0.69-4.67]	0.0
Non-suicidal self-injury					
Non-suicidal self-injury (caregiver report)	35.6%	11.9%	0.00	3.58 [1.72-7.42]	1.5
Psychopathology					
Internalizing symptoms	56.38 (11.35)	51.62 (11.46)	0.03	1.03 [1.00-1.07]	0.0
Externalizing symptoms	54.06 (11.59)	48.62 (11.47)	0.01	1.04 [1.01-1.07]	0.0
Mental health service use	45.0%	71.9%	0.13	1.72 [0.85-3.48]	1.5
Depressive disorders (child report)	21.9%	14.0%	0.26	1.62 [0.70-3.75]	0.0
Anxiety disorders (child report)	28.1%	10.0%	0.00	3.09 [1.43-6.68]	0.0
Bipolar and related disorders (child report)	40.6%	25.3%	0.07	1.94 [0.96-3.93]	0.0
Depressive disorders (caregiver report)	23.8%	9.9%	0.03	2.56 [1.11-5.93]	1.5
Anxiety disorders (caregiver report)	48.1%	40.8%	0.44	1.31 [0.66-2.63]	1.5
Bipolar and related disorders (caregiver report)	14.1%	9.0%	0.42	1.55 [0.54-4.41]	1.5
Obsessive-compulsive and related disorders (caregiver report)	11.9%	14.5%	0.51	0.67 [0.20-2.19]	1.5
Trauma- and stressor-related disorders (caregiver report)	4.7%	7.2%	0.45	0.47 [0.06-3.41]	1.5
Disruptive, impulse-control, and conduct disorders (caregiver report)	43.8%	23.2%	0.01	2.56 [1.11-5.93]	1.5
Temperaments					
Behavioral inhibition	11.75 (3.60)	9.82 (3.72)	0.01	1.14 [1.04-1.26]	0.4
Behavioral activation	1.90 (0.51)	1.63 (0.56)	0.01	2.28 [1.20-4.32]	0.4
Impulsiveness	9.66 (1.48)	8.84 (1.54)	0.01	1.37 [1.10, 1.72]	0.4
Lifestyle factors					
Online social screen time	0.73 (0.74)	0.78 (1.03)	0.81	0.96 [0.67-1.37]	0.4
Screen watch screen time	2.91 (1.42)	2.63 (1.40)	0.29	1.14 [0.89-1.46]	0.4
Video games screen time	1.59 (1.26)	1.54 (1.02)	0.79	1.05 [0.75-1.45]	0.4
Sleep disturbance	40.13 (10.95)	38.74 (11.06)	0.55	1.01 [0.98-1.04]	0.0
Social-environment factors					
Family conflict (child report)	2.56 (2.05)	2.60 (2.34)	0.90	0.99 [0.85-1.15]	0.0
Family conflict (caregiver report)	3.16 (2.03)	2.59 (1.96)	0.14	1.13 [0.96-1.34]	0.0
Parental supervision	4.07 (0.72)	4.22 (0.57)	0.15	0.67 [0.39-1.16]	0.0
Primary caregiver's acceptance	2.64 (0.39)	2.70 (0.36)	0.39	0.68 [0.29-1.62]	0.0
School disengagement	4.47 (1.68)	3.99 (1.56)	0.11	1.18 [0.96-1.46]	0.4
School involvement	12.06 (2.65)	12.23 (2.57)	0.75	0.98 [0.86-1.12]	0.4
Positive view of school environment	18.75 (3.16)	19.19 (3.06)	0.46	0.96 [0.86-1.07]	0.4
Bullying	35.6%	29.1%	0.46	1.31 [0.63-2.73]	2.3
Best friend	65.0%	75.8%	0.19	0.61 [0.29-1.28]	10.3
Friend group	74.1%	88.6%	0.05	0.43 [0.19-0.99]	6.9
Car accident	4.7%	4.1%	0.82	0.79 [0.11-5.79]	0.8
Accident	5.0%	4.1%	0.86	0.83 [0.11-6.11]	0.8
Witnessed domestic violence	23.4%	4.2%	0.00	5.39 [2.33-12.46]	0.8
Sudden death of loved one	22.5%	20.7%	0.86	1.08 [0.47-2.49]	0.8
Parental psychopathology					
Parental trouble with job/fights/police problem	20.6%	15.4%	0.40	1.46 [0.60-3.54]	4.2
Parental nerves/nervous breakdown problem	14.1%	15.1%	0.81	0.88 [0.31-2.51]	3.1
Parental mania problem	9.1%	7.7%	0.65	1.39 [0.33-5.82]	6.5

(Continued)

Table 3. Continued

	Transition from non-Suicidal self-Injury to suicide attempts				Missing (%)
	Transition group (N = 32)	No transition group (N = 229)	P value	HR [95% CI]	
	M (SD)	M (SD)			
Parental drug use problem	18.4%	13.3%	0.43	1.47 [0.57-3.83]	5.0
Parental depression problem	59.1%	40.9%	0.05	2.00 [0.99-4.02]	6.5
Parental alcohol problem	25.9%	18.7%	0.43	1.40 [0.61-3.24]	5.0
Parent hospitalized due to emotional/mental problem	21.9%	12.7%	0.38	1.54 [0.59-3.99]	5.4
Parent been to a doctor or counselor due to emotional/mental problem	49.7%	46.5%	0.81	1.09 [0.54-2.18]	4.2
Parent attempted or completed suicide	9.4%	11.0%	0.69	0.75 [0.18-3.14]	5.4
Perinatal factors					
Maternal medical conditions during pregnancy	0.92 (1.26)	0.63 (0.93)	0.06	1.33 [0.98-1.79]	6.1
Obstetric complications	0.64 (1.02)	0.46 (0.88)	0.29	1.21 [0.85-1.72]	9.2
Birth weight	6.69 (4.39)	7.27 (4.65)	0.47	0.97 [0.90-1.05]	5.7
Paternal age at child's birth	30.88 (7.83)	31.76 (6.60)	0.43	0.98 [0.92-1.03]	5.0
Maternal age at child's birth	28.16 (6.23)	30.22 (5.84)	0.07	0.95 [0.89-1.00]	1.9
Planned pregnancy	51.6%	66.2%	0.11	0.57 [0.28-1.14]	2.7
Medication use during pregnancy	13.1%	18.6%	0.39	0.59 [0.18-1.95]	7.7
Maternal alcohol/substance use during pregnancy	20.3%	9.0%	0.05	2.47 [1.01-5.99]	2.3
Premature birth	15.6%	24.3%	0.30	0.60 [0.23-1.57]	0.4

findings.^{11,20} Specifically, higher severity of suicidal ideation, as coded in the current study, indicates a progressively higher intention and planning for suicide, which is an important volitional factor linking suicidal ideation to the enactment of suicide attempts.¹¹ Overall, these findings not only provide empirical support for the ideation-to-action framework in a pediatric sample but also suggest that preventive interventions may benefit from targeting suicide capacity and volitional factors.

In addition to established risk factors, we identified several novel predictors of suicide attempts in children experiencing suicidal ideation, including increased engagement with social media and diminished parental supervision. Although these factors have been associated with suicide attempts among the general population, our study is the first to demonstrate their prospective

association with future suicide attempts in this high-risk group. Specifically, our finding on the association between online social screen use and transition to suicide attempts aligned with previous studies that identified an independent, direct link between heavy social media or internet use and increased suicide attempts.⁵⁵ Possible pathways include exposure to suicide-related content or withdrawal from the real-world through heavy use of social media, but these are beyond the scope of the current study and require future investigation. Regardless of the underlying mechanisms, our finding highlights the potential value of prevention and intervention strategies targeting maladaptive screen usage. Additionally, our analyses found that parental supervision might play a pivotal role in reducing the likelihood of the transition, as supported by the research of King et al.⁵⁶

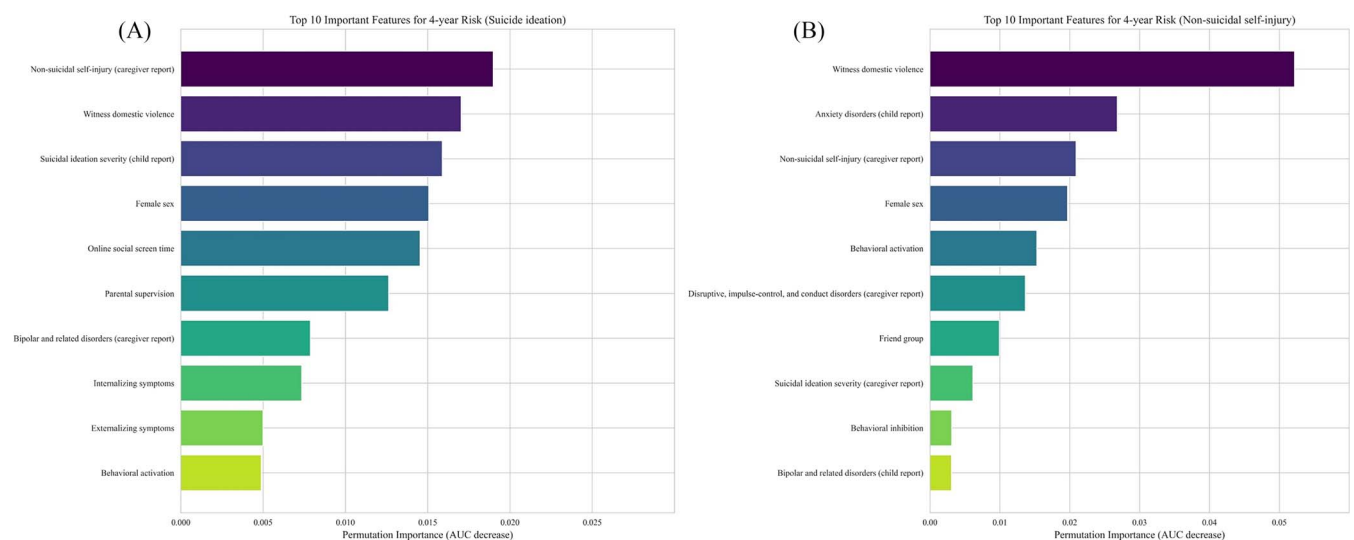


Figure 3. Predictor importance for models predicting transitions to suicide attempts. (A) Predictor importance for the model predicting the transition from suicidal ideation to suicide attempt. (B) Predictor importance for the model predicting the transition from non-suicidal self-injury to suicide attempt.

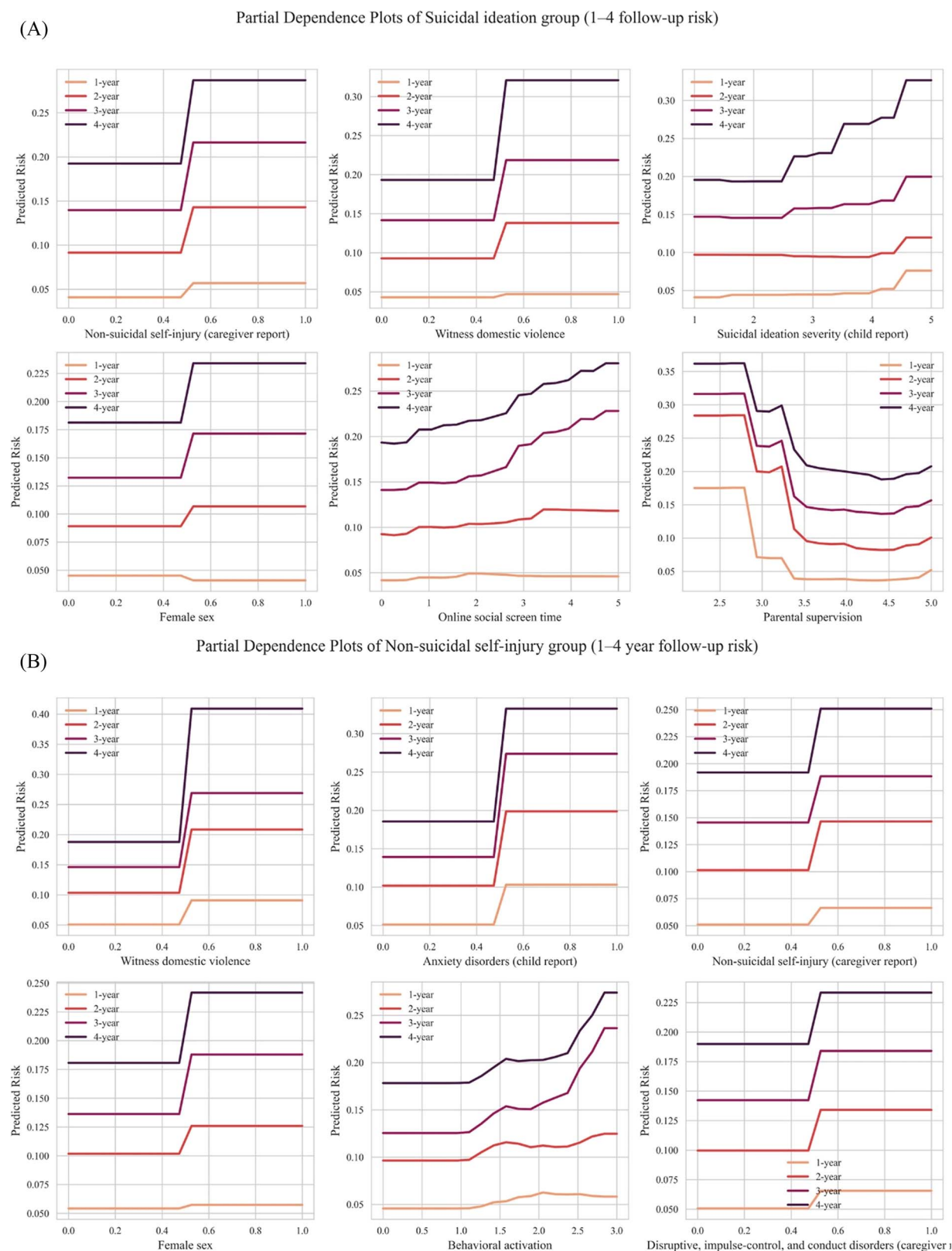


Figure 4. Partial dependence plots for the most influential predictors. (A) Plots for the most influential predictors for the model predicting the transition from suicidal ideation to suicide attempt. (B) Plots for the most influential predictors for the model predicting the transition from NSSI to suicide attempt.

Enhanced supervision from caregivers may facilitate early detection and intervention in cases of suicidal ideation, thereby reducing the progression to suicide attempts in children. This

observation emphasizes the critical role of familial engagement in preventing the transition from suicidal ideation to suicide attempts.

Similarly, among children engaging in NSSI, comorbid anxiety disorders as well as disruptive, impulse-control, and conduct disorders emerged as important predictors of subsequent suicide attempts. Children with NSSI who also present with comorbid anxiety disorders may experience heightened emotional distress and impaired emotion regulation, increasing their vulnerability to suicide attempts. Likewise, the presence of disruptive, impulse-control, and conduct disorders increases vulnerabilities possibly through elevated behavioral activation, contributing to acting on suicidal urges. Although the mechanisms underlying these associations remain to be clarified, they point to important targets for future investigation. These findings underscore the importance of vigilant monitoring of children's psychopathology to prevent the progression from NSSI to suicide attempts.

Our findings should be interpreted in light of their limitations. First, this is an observational study, which, despite being prospective, cannot establish causality. Additionally, the predominantly White participant cohort may limit the generalizability of our findings across diverse racial, ethnic, and cultural groups. Moreover, our study used a preliminary variable screening step prior to model development. This approach was intended to reduce noise and mitigate overfitting given the large number of predictors and relatively small number of outcome events. However, it may have led to the omission of some potentially informative variables that did not show initial bivariate associations. Future studies with larger samples will be important to evaluate whether our findings remain robust without such pre-screening. Furthermore, our analysis did not include specific measures to assess practical capacity, often associated with access to lethal means. While prior studies emphasize the role of firearms, suicide in children and young adolescents often involve methods like hanging, which accounts for approximately 70% of suicides among 10- to 14-year-olds globally.⁵⁷ The widespread availability of such means complicates the assessment of practical capacity, presenting a significant methodological challenge.

Despite its limitations, our study brings significant contributions to the field. Leveraging the ABCD study's large, nationally representative sample and comprehensive risk factor assessments, we robustly identified predictors of suicide attempts among children with suicidal ideation or NSSI. Our research is pioneering in its focus on the initiation of suicide attempts during the pivotal developmental window of late childhood—a period marked by sharp increases in suicide risk. Furthermore, the population-based sample lends considerable credence to our findings. Additionally, the use of machine learning allowed us to capture complex, non-linear patterns that traditional approaches are less suited to detect. What is more, using diagnostic interviews to assess and distinguish initiation from reoccurrence greatly improved the accuracy and precision of clinical status judgment. Finally, our comprehensive examination of the transition from suicidal ideation and from NSSI to suicide attempts enhanced the nuanced understanding of suicide risk factors, potentially informing selective prevention in suicidal ideation and NSSI groups.

In conclusion, our study delineated key risk factors for the initial progression to suicide attempts among children with suicidal ideation or NSSI, respectively. Derived from these risk factors, our predictive models demonstrated notable accuracy in identifying children at elevated risk for their first suicide attempt. Consequently, our findings may pave the way for the development of targeted strategies, especially an integrated child- family intervention approach, to bolster prevention in the realm of child mental health.

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Supplementary material

Supplementary material is available at the *American Journal of Epidemiology* online.

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Conflict of interest

The authors declare no conflicts of interest.

Disclaimer

The views expressed in this article are solely those of the authors and do not represent the views of the NIH or the ABCD consortium.

Data availability

Data from the Adolescent Brain and Cognitive Development (ABCD) Study (<https://abcdstudy.org>) were used in this research. Access to the data is available to qualified researchers upon application.

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